

PROJECT DOCUMENT

FoodFresh AI

Know Your Food. Save Your Food. Waste Less.

An AI-assisted food management prototype connecting visual food analysis with pantry decisions.

JUDGE-FACING TECHNICAL AND SOCIAL-IMPACT OVERVIEW

Project state reviewed 30 September 2026

Contents

01. Executive summary
02. Problem statement
03. Proposed solution
04. Features and status
05. Objectives
06. Literature survey
07. Research gaps
08. Technical concepts
09. Machine-learning concepts
10. Datasets and references
11. Data preparation
12. Models and rules
13. Hybrid architecture
14. SQLite and persistence
15. FreshoBuddy
16. Eat First
17. Limitations
18. Future scope
19. Social impact
20. Feasibility
21. Cost estimation
22. Scalability
23. Security and privacy
24. Project learning
25. Project contribution
26. Judge-focused points
27. Future vision
28. Main motto
29. References

Implementation status, research candidates and limitations are labeled separately. Project test scores refer to the named local test split; external model-card scores are identified as external.

1. Executive summary

FoodFresh AI is a prototype food-management platform that combines image-based food identification, visual freshness classification, reference-based shelf-life estimates, pantry history and consumption-priority guidance. Its product goal is to help a household notice food it already has and decide what to use first.

The problem is practical: groceries are stored in multiple places, conditions differ by food, and people may forget what they bought. A date reminder or inventory list can record an item, but it does not necessarily connect its visible condition and storage context to a next step. FoodFresh AI brings those pieces into one workflow.

The current backend runs a hybrid vision pipeline using Grounding DINO for open-vocabulary object detection, SigLIP 2 for image/text semantic matching, and a ResNet-50 raw-food specialist. The fused food identity is passed to an AgriFreshNET fine-tuned ResNet-18 freshness model and a deterministic USDA FoodKeeper reference/estimation service. Pantry records and chat history persist in SQLite; an explicit Eat First rule engine ranks saved items. FreshoBuddy provides a conversational interface with structured pantry answers handled deterministically and other food questions eligible for a backend Gemini response when configured.

Interpretation boundary. Image analysis estimates visible surface condition. It cannot inspect internal contamination, measure microbes, smell food, or certify safety. FoodKeeper durations and remaining-life values are estimates that depend on the correctness of food identity, dates and storage context.

Food image	»	Detect and identify	»	Visible freshness	»	FoodKeeper estimate	»	Pantry and Eat First	»	User action
------------	---	---------------------	---	-------------------	---	---------------------	---	----------------------	---	-------------

FoodFresh AI is differentiated by this system-level connection, rather than by claiming a new neural architecture. Its present contribution is turning multiple predictions and reference rules into a persistent household workflow. Recipes from pantry ingredients, NGO donation, impact measurement and a full PWA remain future scope.

2. Problem statement

Households often lack a reliable, low-effort way to track food identity, storage context, visible deterioration and intended use together. Items can be forgotten in a refrigerator or pantry; foods differ in storage guidance; and appearance is not always easy to assess consistently. Static expiry reminders do not observe current appearance, while a food classifier by itself does not tell a household what to do next.

This project addresses the design problem: *How can an accessible digital system combine a food photograph, user-provided storage context and a household pantry record to support more timely food-use decisions without overstating what computer vision can know?*

The social importance is substantial, but the project makes no claim that the prototype has already reduced waste. UNEP's 2024 report estimates 1.05 billion tonnes of food waste at retail, food-service and household levels in 2022; households contributed about 60% of that total [8]. The estimate motivates prevention tools; it is not an impact metric for this project.

Food waste prevention can conserve the land, water, energy and labor used to produce food. FoodFresh AI targets one small point in that chain: the household decision between use soon, store appropriately, or seek another safe and practical action.

3. Proposed solution

Observe » Understand » Predict » Recommend » Act

1. **Observe:** the user submits a food image and storage information.
2. **Understand:** computer-vision components propose an identity; a fusion step resolves model evidence and may return uncertainty.
3. **Predict:** the freshness classifier estimates a visible class, while FoodKeeper rules estimate a storage-life window for a canonical food/storage match.
4. **Recommend:** the application combines that window, freshness and saved pantry state to prioritize items.
5. **Act:** the user can inspect the result, save it to pantry history and ask FreshoBuddy for guidance.

Prediction is a model or rule output. **Recommendation** is a ranked next step, such as use soon. **Action** is something the user actually does-cook, store, share or discard. The prototype supports analysis and prioritization; it does not yet complete meal generation or donation transactions.

Example future workflow: “Spinach appears to have a short estimated quality window » use it soon » show recipes based on spinach, tomato and onion » let the user choose a meal.” The recipe step is proposed future work; it is not represented as an implemented pantry-aware recipe feature.

4. Features and implementation status

Feature	How it works	User benefit	Status
Food image upload	Image is sent to the authenticated FastAPI analysis route.	Begins a visual analysis without manual class selection.	Implemented
Camera-based capture	Current workflow accepts an image; direct in-app live camera capture is not verified as a distinct capability.	Could reduce steps on mobile.	Not verified / future enhancement
Food recognition	Grounding DINO detection, SigLIP 2 matching and raw-food ResNet-50 evidence are fused.	Provides a candidate food identity.	Implemented
Freshness assessment	Active AgriFreshNET fine-tuned ResNet-18 classifies Fresh, Rotten or Slightly Spoiled; confidence gates can mark uncertainty.	Highlights visible changes that merit attention.	Implemented; surface appearance only
Shelf-life estimation	FoodKeeper storage references and application rules combine food match, storage type and days stored.	Offers a contextual reference window.	Implemented as estimate, not ML regression
Pantry and analysis history	Authenticated item analyses are stored in SQLite and can be reloaded.	Reduces reliance on memory.	Implemented
Dynamic remaining-days display	Stored analysis dates and the estimated window drive a changing countdown/remaining value.	Makes time context visible.	Implemented; depends on estimate and user context
Eat First prioritization	Deterministic tier rules rank records by visible freshness and remaining estimated window.	Turns records into a use-order suggestion.	Implemented
FreshoBuddy	Pantry intents use backend-authoritative, deterministic answers; other conversational food guidance can use Gemini when available.	Lets users ask questions in natural language.	Implemented; external model availability/configuration applies
Authentication	Registration, login, expiring sessions and per-user access checks guard saved data.	Separates household histories.	Implemented
ML feedback	Feedback route and SQLite table are present for future error collection.	Creates a path to review model mistakes.	Infrastructure present; zero records in last verified DB
Progressive Web App	Install manifest, offline cache and push-notification package are not present/verified.	Could support install and limited connectivity.	Future scope
Pantry-aware recipe recommendations	No structured recipe generator using pantry priority is implemented.	Could convert “use soon” into a meal choice.	Future scope
Donation, waste-risk card, notifications and impact analytics	No verified NGO partnership or end-to-end donation/measurement flow exists.	Could redirect safe surplus and measure outcomes.	Future scope

5. Project objectives

Technical objectives

- Identify common food items from images while exposing uncertainty when evidence conflicts.
- Estimate visible freshness condition using a trained image classifier.
- Combine a food identity with reference storage guidance and user context to estimate a quality window.
- Maintain per-user pantry records and history in persistent storage.
- Rank saved items with explainable, deterministic urgency rules.
- Provide natural-language support without allowing generated text to overwrite structured pantry values.
- Evaluate models on held-out data and representative image fixtures, keeping dataset and external model-card scores clearly separated.

Social and environmental objectives

- Encourage people to notice and use food they have already purchased.
- Promote more deliberate storage and household inventory habits.
- Build a future pathway from “use soon” to recipe or verified food-rescue options.
- Define impact measures before making a waste-reduction claim.

6. Literature survey

Research/source	Approach and contribution	Gap relevant here
Mureşan & Oltean (2018), Fruits-360 [1]	Introduces a rotating-produce image dataset and CNN experiments for fruit recognition. It is useful for controlled visual classification.	Rotating isolated objects on a white background do not reproduce cluttered kitchens, mixed produce, spoilage diversity or storage history.
Risdin et al. (2026), AgriFreshNET [2]	Dataset record describes RGB produce images with Fresh, Semi-Fresh and Rotten stages and approximate post-harvest interval annotations.	Taxonomy-level intervals are not longitudinal, per-image remaining-life measurements with measured storage conditions.
Liu et al. (2023/2024), Grounding DINO [3]	Combines a Transformer object detector with grounded language information for open-set detection and text-conditioned localization.	General object detection does not guarantee correct food identity or safe freshness conclusions; task-specific validation remains necessary.
Tschannen et al. (2025), SigLIP 2 [4]	Vision-language encoders trained with a sigmoid contrastive objective and additional representation-learning techniques; supports image/text matching.	General semantic compatibility is not a calibrated probability of correct identity for every produce domain.
Ma et al. (2016), computer vision in agri-food quality [5]	Reviews imaging methods and their uses in food-quality assessment.	Quality measurement needs ground truth and often more than RGB appearance; an application must state what its sensors do not observe.
Palumbo et al. (2023), conventional imaging review [6]	Reviews conventional computer-vision systems used to assess produce quality, external defects and selected internal attributes.	Industrial systems may use controlled illumination, cameras, or measurements not available in a household phone image.
UNEP Food Waste Index (2024) [8]	Provides global measurement methodology and estimated 2022 retail, food-service and household waste totals.	Global estimates establish motivation, not a measured outcome for this prototype.

The research points to a practical integration gap: recognition, freshness measurement, storage references and household actions are usually separate tasks. FoodFresh AI explores how to connect them in a usable workflow. It does not claim a novel neural architecture or a validated food-safety instrument.

7. Research gaps and project response

Existing concept	Relevant limitation	FoodFresh AI response
Food recognition	A name alone does not tell the user what to use first.	Feeds an analysis record, storage estimate and pantry priority.
Freshness classifier	Class output may not include elapsed age, storage history or an action.	Combines visible class with FoodKeeper and pantry context; keeps rules explainable.
Controlled image datasets	Clean background and uniform capture can differ from phone photos.	Uses AgriFreshNET for freshness work and checks selected real-world fixtures; external validation remains needed.
Static storage tables	Reference durations may not reflect a particular item's actual history.	Accepts storage type and days stored and labels results estimates.
Assistant-generated advice	Free-form text can invent foods or contradict calculated values.	Handles eat-first and "why?" with deterministic pantry-backed replies; routes other questions through a constrained context-aware assistant.
Food waste action	Prediction may stop before user does anything.	Currently provides pantry prioritization; recipes, donation and measured outcomes are planned rather than claimed.

These are design gaps observed across component boundaries, not a claim that every existing food-management product has each limitation.

8. Technical concepts used

Concept	What it is / why it is used	Use in this project
Computer vision	Methods that extract information from images.	Detects and classifies food appearance from a submitted photograph.
Image classification	Assigns a label to an image or crop.	Freshness ResNet-18 produces one of three visible-condition classes.
Object detection	Finds objects and bounding boxes within an image.	Grounding DINO proposes food and scene regions before classification.
Open-vocabulary detection	Uses text prompts to detect concepts beyond a fixed label set.	Grounding DINO receives language-conditioned object concepts; this does not ensure all food items are in coverage.
Vision-language model	Encodes image and text in a related representation space.	SigLIP 2 scores candidate food descriptions for semantic matching.
Feature embedding and similarity	Numerical vectors represent content; similarity compares vectors.	Supports ranking candidate image/text matches.
Model fusion	Combines outputs from multiple components.	Deterministic logic reconciles specialist, semantic and detection evidence; disagreement can produce uncertainty.
Transfer learning	Reuses learned features from an existing model.	Freshness V2 is a locally fine-tuned ResNet-18; V3 candidate initializes EfficientNet-B0.
Confidence and uncertainty	Scores express model preference; a threshold can decline to assert a class.	Freshness probabilities use temperature calibration and uncertainty gates. Scores are not food-safety probabilities.
Image preprocessing	Transforms pixels into the form expected at inference.	RGB decoding, EXIF orientation, resize/crop and ImageNet normalization must match training.
REST API / FastAPI	HTTP resources allow frontend and backend services to communicate.	FastAPI serves auth, analysis, history and assistant routes; React sends requests and renders responses.
SQLite	Embedded relational database stored in a local file.	Persists users, sessions, pantry/history, conversations and messages for a single-instance prototype.
Authentication	Verifies identity and authorizes requests.	PBKDF2 password hashing and expiring bearer sessions protect account-scoped API data.
Dynamic countdown	Computes a changing remaining value from a stored date/window.	Pantry views derive days remaining from saved analysis and estimate fields.
Assistant architecture	Combines structured application state with a language model.	Pantry triage is deterministic; Gemini is an optional conversational layer for other questions.
PWA	Web-app manifest, service worker and install/offline patterns.	Discussed as future scope; a full PWA capability is not verified in the current frontend.

9. Machine-learning concepts

Classification learns a mapping from input to a finite set of labels. **Multi-class classification** selects among more than two labels. Freshness uses Fresh, Rotten and Slightly Spoiled; food recognition proposes an identity from its own vocabulary. These tasks are distinct: knowing an item is a tomato does not determine its freshness.

Pretraining supplies general visual features learned from other data. **Fine-tuning** updates some or all of those parameters on the target dataset. Augmentation-such as modest rotation or horizontal flip-can expose a model to plausible input changes. It cannot create missing food classes or real storage histories.

Data are separated into **training** (fit parameters), **validation** (select settings/checkpoints), and **test** (final held-out measurement). Leakage occurs when near-duplicate or related source images cross splits, making scores too optimistic. The local AgriFreshNET manifest was checked for shared base stems across splits.

Metric	Meaning	Why it matters
Accuracy	Correct predictions divided by all examples.	Easy summary, but can hide poor minority-class performance.
Precision	Of predictions for a class, the share correct.	Shows false alarms for that class.
Recall	Of actual examples in a class, the share found.	Shows misses for that class.
F1-score	Harmonic mean of precision and recall.	Summarizes trade-off; macro F1 weights classes equally.
Confusion matrix	Counts actual versus predicted classes.	Shows which freshness stages are confused.
ECE / NLL	Calibration/error measures for probability confidence.	Help assess confidence quality; do not prove safety or generalization.

Domain shift occurs when deployment images differ from training data in food varieties, lighting, background, camera, occlusion or spoilage. A model can score strongly on a held-out split from the same collection process and still fail on unfamiliar phone images. The project's 11 selected fixtures illustrate why accuracy must be measured on broader, independent data; they are not a representative benchmark.

10. Datasets and data sources

Source	Contents and size	Use and role	Limitations
Fruits-360 [1, 9]	The project's local V2 preparation report records 128,210 images in 24 selected classes (76,961 train / 19,233 validation / 32,016 test), with zero path duplicates or split leaks. The original 2018 paper described an earlier 90,483-image, 131-class snapshot; the current upstream original-size repository is versioned and changes over time (110,482 images / 153 classes at the source revision checked 2026-09-30). These figures refer to distinct snapshots and project preparations.	Controlled produce recognition benchmark and preparation/evaluation source; not freshness ground truth. The current API's active recognizer is the hybrid pretrained pipeline.	Rotated, isolated produce on a processed white background is unlike cluttered household photographs; old training and latest upstream stats are not interchangeable.
AgriFreshNET [2]	Published dataset record: 14,500 RGB images, 24 food/stage classes, Fresh/Semi-Fresh/Rotten, and approximate taxonomy-level post-harvest intervals. Project-local prepared subset: 14,160 images, 8 foods x 3 stages (24 folders), 512 x 512 RGB; 9,856 train / 2,131 validation / 2,173 test.	Local freshness V2/V3 training and test evaluation; interval-class research candidate. Local classes are Banana, Bittermelon, Cucumber, Eggplant, Orange, Papaya, Pineapple and Tomato.	Local subset is not the full published dataset. Published interval annotations are approximate category ranges, not observed per-image elapsed or remaining time with storage sensors/conditions.
USDA FoodKeeper [7, 10]	USDA FSIS-developed consumer storage guidance. The project's local JSON index contains 661 product entries across 25 categories per project audit.	Reference/knowledge source for storage-dependent duration lookup and explanatory tips. It is not a trained ML model.	Generic guidance cannot reconstruct an individual item's actual harvest, temperature history, handling, package state or safety.
Raw Food Recognition [11]	External Hugging Face repository: 90 single-class food categories. The model card reports 97.84% best validation accuracy for its ResNet-50 on its source dataset.	Pretrained raw-food specialist in the hybrid recognition pipeline. The model-card result is external, not a FoodFresh AI benchmark.	Class coverage, image distribution and metrics are those of the external project; performance is not guaranteed on FoodFresh AI photos.
Real-world regression fixtures	11 selected labeled local images were used for a compact smoke check (e.g., tomato, orange, chilli, bread).	Useful for checking obvious regressions and API inference.	Small and selected; not statistically representative. Chilli is absent from the local AgriFreshNET training classes; dried red chilli had no image fixture.

Published versus local data. The AgriFreshNET publication's 14,500/24 figures are kept separate from the local 14,160-image, eight-food prepared subset. Likewise, Fruits-360 figures are version-specific; no one number should be treated as a timeless property of every release.

11. Data cleaning and preprocessing

- Manifest rows are organized with food identity, freshness label, image path and the source folder's interval annotation.
- All local manifest paths were verified to resolve; the train, validation and test split totals are 9,856, 2,131 and 2,173.
- Grouping by base image stem found no stem crossing between splits. This reduces one form of near-duplicate leakage; it does not prove independence across every capture session or source.
- Image decoding applies EXIF orientation correction and RGB conversion. Freshness evaluation uses the training-compatible resize/crop and ImageNet normalization.
- Freshness V3's evaluation transform is resize to 256, center crop to 224, tensor conversion and ImageNet normalization. Training uses limited image augmentation as specified in the trainer.
- The classifier is evaluated on the test manifest after model/checkpoint selection on validation data.

The local AgriFreshNET analysis found 14,160 images distributed evenly among freshness stages, with 590 samples in each food-stage folder. Equal counts are useful for controlled comparison, but do not represent real-world prevalence. Images with different lighting, occlusion, packaging or multiple items can still produce difficult predictions.

12. Models, rules and measured results

Component	Role/output	Project status and evidence
Grounding DINO Base [3]	Text-conditioned object detection and candidate food regions.	Active in the hybrid API pipeline; not fine-tuned by this project.
SigLIP 2 Base Patch16-224 [4]	Image/text semantic ranking for food concepts.	Active pretrained component; no project-specific independent benchmark is claimed.
Raw-food ResNet-50 [11]	Specialist single-class raw food predictions.	Active pretrained component; 97.84% is the external model-card validation result only.
Freshness ResNet-18 V2	Fresh / Rotten / Slightly Spoiled visible-surface classification.	Active. On the local AgriFreshNET 2,173-image test split: 96.04% accuracy, 0.9605 macro F1, 0.9604 balanced accuracy. Calibration temperature 1.1118 is fit on validation logits; test ECE improved 0.0231 » 0.0196 and NLL 0.1523 » 0.1426.
Freshness EfficientNet-B0 V3	Research candidate with three freshness classes.	Not promoted. Independent held-out evaluation: 82.60% accuracy / 0.8212 macro F1. The trainer reported 93.65% / 0.9357, but the result was not reproduced using the saved checkpoint and same split; the mismatch remains unresolved.
USDA FoodKeeper logic	Canonical food and storage lookup plus remaining-window rules.	Active reference/rule system. Not an ML shelf-life model.
24-class interval classifier	Image-to-food/stage taxonomy interval research experiment.	Unintegrated candidate. 94.29% interval-class test accuracy, macro F1 0.9430. This is not day-level regression or observed shelf-life performance.
Eat First engine	Priority tiers with reasons.	Active deterministic rules, not ML.
Gemini API	Conversational response generation for non-deterministic assistant intents.	Backend integration exists; requires server-side key/network/model availability. Not needed for deterministic pantry triage.

The V2 score is a held-out result from the local prepared dataset, not evidence that all crops, cameras or household conditions achieve that accuracy. The small fixture smoke check had V2 correct on 7 of 11 images and included errors on potato, apple, pomegranate and bread. The fixture set is too small for a generalized performance claim.

13. Hybrid AI architecture

React image form	»	FastAPI validation and image decode	»	Grounding DINO regions
------------------	---	-------------------------------------	---	------------------------

v candidate crop / image context

Raw-food ResNet-50	+	SigLIP 2 image/text scores	»	Evidence fusion and uncertainty	»	Food identity
--------------------	---	----------------------------	---	---------------------------------	---	---------------

v parallel downstream assessment

Freshness V2	+	FoodKeeper reference + storage/days	»	Analysis result	»	SQLite pantry/history
--------------	---	-------------------------------------	---	-----------------	---	-----------------------

v persistent pantry context

Deterministic Eat First	»	FreshoBuddy / user decision
-------------------------	---	-----------------------------

The pipeline is modular: detector output provides regions; the semantic and specialist classifiers provide complementary identity evidence; deterministic fusion produces an identity or uncertainty; freshness and shelf-life components provide separate outputs. A food identity match is still fallible, and downstream values inherit that uncertainty.

REST endpoints connect the browser and backend. The health endpoint reports the active freshness checkpoint and calibration configuration. Research candidates remain outside production routing.

14. Database - SQLite

SQLite is appropriate for a local demonstration and a single backend instance: it is embedded, requires no separate database server, and makes persistence easy to inspect and back up. It is not assumed to be the final choice for high concurrency or multi-instance deployment.

The verified database contains seven tables: `users`, `auth_sessions`, `pantry_history`, `pantry_meta`, `conversations`, `messages` and `ml_feedback`. Pantry fields record analysis identifiers, food/form, freshness, estimate inputs/results, storage information and model-version provenance. Conversations and messages are user-scoped.

At the last read-only verification for this document, the database held 37 pantry records, 6 users, 12 conversations, 34 messages and 0 feedback records. These are development-instance counts, not product adoption statistics. No user database file is required in a public source checkout.

React client	<- >	FastAPI authenticated routes	<- >	SQLite: users · pantry · chat · feedback	>>	Dashboard and history
--------------	---------	------------------------------	---------	--	----	-----------------------

15. FreshoBuddy AI

FreshoBuddy is the conversational food companion. It can respond to household questions using saved pantry context and can provide general food-management guidance. The service loads its Gemini key on the backend; React does not need to receive the provider secret.

For “Which food should I eat first?” and a follow-up “Why?”, the implementation uses deterministic pantry triage. It computes from the application's records and explains the saved remaining-day/urgency values. This avoids asking a generative model to recalculate structured pantry facts. Other intents can use Gemini with contextual data when configured; generated text is not an authoritative food-safety inspection.

Conversation records and messages persist in SQLite and are scoped to the authenticated user. If external model access is unavailable, the experience depends on the implemented deterministic/fallback behavior. Exact availability depends on local secret configuration and network conditions.

16. Eat First system

Eat First is a four-tier deterministic priority engine. It reads the FoodKeeper remaining estimate and the freshness label. It does not train on user history, forecast future spoilage, or use a neural ranking model.

Tier	Rule outline	Example explanation
VERY_HIGH	Rotten label, no estimated days left, or an estimated window at or below about one day.	Quality estimate is expired/critically short; a rotten visible label triggers caution.
HIGH	Slightly spoiled or estimated average/maximum window at or below about three days.	Use soon because the estimated window is short.
MEDIUM	Estimated remaining window at or below about seven days.	Prioritize within the week.
LOW	Longer estimate with fresh visible class.	Ample estimated window remains.

Ties are handled as ties; the assistant may explain both items. If the FoodKeeper match is unavailable, the system returns unavailable guidance instead of inventing a duration. The priority is a decision aid, not an instruction to eat food that smells, feels or appears unsafe.

17. Current system limitations

- **Image dependence:** blur, glare, shadows, distance, occlusion, mixed items and unusual crops can change predictions.
- **Visible-condition boundary:** an RGB image cannot detect internal deterioration, microbes, allergens, toxins or unsafe temperature history.
- **Class coverage:** the local AgriFreshNET freshness subset includes eight foods. Broader pretrained identity models do not make the freshness classifier universal. Chilli coverage is especially limited in freshness training; bread is outside the AgriFreshNET classes.
- **Food identity errors:** the FoodKeeper lookup depends on a correct canonical match. Ambiguous identity can make shelf-life estimates unavailable or misleading.
- **Reference-window uncertainty:** FoodKeeper guidance is general; actual freshness depends on initial quality, package, ripeness, time and temperature history.
- **Dataset shift:** same-source held-out data is not equivalent to new households, cameras, regions or produce varieties.
- **Freshness V3 discrepancy:** trainer-reported metrics do not match independently reproduced checkpoint evaluation. V3 was not promoted.
- **Shelf-life ground truth:** taxonomy intervals are not measured per-image remaining-life observations. The interval candidate does not estimate day errors.
- **Evaluation:** the 11-image smoke set is small and selected; no broad external clinical, microbiological or prospective household trial has been performed.
- **Prototype operations:** SQLite is single-instance oriented, Gemini relies on external configuration/connectivity, and no verified NGO or PWA rollout exists.

Food safety: A “Fresh” prediction is not a safety guarantee. Users should follow applicable food-safety guidance and their own sensory checks; discard food if it appears unsafe.

18. Future scope

Future feature	Concept and value	Requirements before release
Pantry-aware recipes	Use pantry inventory, preferences and urgency to recommend a practical recipe. This turns “use soon” into a meal option.	Recipe sources/attribution, allergen and preference handling, verified ingredient substitutions, user choice.
Installable PWA	Manifest, service worker, install flow, limited offline pantry access and browser camera capture.	Offline privacy/sync design, permission UX, conflict resolution and device testing.
Food-waste risk card and alerts	Notify a user when saved items approach an estimated quality window.	Opt-in notifications, quiet hours, timezone/date correctness and false-alarm controls.
Verified donation pathway	Let a user ask about redirecting suitable surplus through a verified food-rescue organization.	Actual local partners, accepted-food rules, food-safety logistics, pickup and legal responsibilities. No partnership is currently claimed.
Impact analytics	Track weighed food consumed, donated or discarded; calculate estimates only from recorded evidence.	Consent, definitions, baseline method and recognized conversion factors. No impact figure should be inferred from predictions alone.
Broader freshness/shelf-life learning	Collect consented longitudinal observations with timestamps, storage temperature/humidity and outcomes.	Careful sampling, privacy, ground-truth protocol and external validation before regression claims.
Feedback loop	Collect user corrections, review them, and curate labeled errors for later evaluation.	Moderation, data governance, consent, duplicate checking and held-out evaluation rules.
Regional and multilingual coverage	Support local food names, varieties and storage practices.	Regional dataset and domain experts; translation alone is not model coverage.

19. Social and environmental impact

Food waste has economic, social and environmental consequences. UNEP estimates that 1.05 billion tonnes were wasted at consumer-facing stages in 2022, including inedible parts; 60% was attributed to households [8]. FoodFresh AI aligns with the intent of SDG target 12.3 to reduce food waste, but this prototype has not measured household behavior or avoided waste.

The hypothesized pathway is:

Better awareness » better timing decisions » more food used or redirected » less avoidable waste » better use of production resources

Each arrow requires evidence. A dashboard may show an estimate; a credible impact evaluation needs a baseline, documented actions and preferably measured quantities. A predicted “saved” item is not necessarily eaten, donated or otherwise kept out of waste.

20. Feasibility

Dimension	Prototype assessment	Conditions for broader deployment
Technical	Feasible as a local full-stack prototype: React/Vite, FastAPI, pretrained vision services, freshness inference and SQLite are integrated.	Benchmark latency and model memory; improve coverage, monitoring, calibration and operational fallbacks.
Operational	Suitable for a demonstration with a local server and explicitly managed API keys.	Reliable hosting, dataset/model versioning, support, incident response and data-retention policy.
Economic	Low incremental prototype cost when existing hardware is used; cloud inference and Gemini calls create variable expense.	Measure per-image compute, request volumes, image storage, API usage and service-level requirements.
Social	Addresses a familiar household problem through understandable guidance.	Usability study, accessibility/language testing, culturally appropriate food coverage and food-safety review.

21. Cost estimation

No invoices or usage-based cloud/API cost study are available in the project records. Values below are planning bands only, not vendor quotations or current price guarantees. They assume Indian rupees, existing developer hardware, a student demonstration, low traffic and no paid labor.

Scenario / line item	Planning allowance	Assumptions and excluded costs
Local prototype hardware/software	■0 incremental to about ■5,000 for accessories/storage	Existing laptop/GPU, free/open-source software, local SQLite; excludes labor, electricity and equipment purchase.
Optional domain	■700-■2,000 per year (rough planning allowance)	Registrar and domain ending vary; renewals, tax and privacy protection may add cost.
Hosted classroom demo	Budget only after selecting provider; a provisional ■1,000-■10,000/month may cover small CPU hosting and modest storage.	GPU inference, Gemini usage, large egress, backups, taxes and production availability are excluded; this is not a price quote.
Production deployment	Not responsibly estimable from current evidence.	Requires region, images/month, GPU/CPU choice, concurrent load, retention, security, uptime, support and donation integration scope.

Future budgeting should measure the number and size of images, average inference time, model memory, LLM tokens per conversation, storage retention and backup frequency. Hosted model inference, database, object storage, monitoring, domain, PWA distribution and NGO operations are separate cost drivers.

22. Scalability

- **Models:** broaden produce classes using licensed, regionally representative data and evaluate each new class outside the training source.
- **Inference:** benchmark CPU/GPU options, use model-serving processes with explicit resource limits, and separate long-running inference from interactive web processes.
- **API:** package backend components consistently, add health/latency/error monitoring and scale stateless API workers after moving state out of one local file.
- **Database:** retain SQLite for small local demos; migrate to managed PostgreSQL when concurrency, multiple application replicas, backup or transactional needs exceed a single-file model.
- **Images:** use object storage with retention controls rather than putting image binaries into relational tables; collect only what is needed.
- **Product:** add multilingual names, regional FoodKeeper mapping, preferences and verified food-rescue partners only after data and operating rules are established.

23. Security and privacy

- Passwords are stored as salted PBKDF2-HMAC-SHA256 hashes; authentication uses expiring bearer sessions.
- Routes resolve the user from the session and scope pantry/conversation records to that user. Access isolation should remain part of regression testing.
- Gemini credentials are read by backend code from environment configuration and must never be committed or exposed in frontend bundles.
- Images and pantry data can reveal household routines. A deployment needs a clear retention/deletion policy, consent for model feedback and secure backups.
- SQLite database files, raw datasets, model checkpoints, environment files and large archives are excluded from the source push; users should create their own local data/configuration.
- No security certification, independent penetration test or formal privacy audit is claimed.

24. Project learning

- **Technical:** image classification/detection, transfer learning, preprocessing consistency, REST integration, database persistence and full-stack request flows.
- **Research:** dataset provenance, split leakage checks, class coverage, calibration, model uncertainty and the difference between dataset test accuracy and real-world behavior.
- **System design:** multiple models need an explicit fusion policy; reference rules must not be mislabeled as learned models; generated language should not overrule structured user records.
- **Product:** a prediction becomes useful only when it supports an understandable next step. That action still depends on user judgment and reliable system context.
- **Responsible use:** visible appearance is not food safety, taxonomy intervals are not observed days, and external model-card performance is not a project benchmark.

25. Project contribution

FoodFresh AI's contribution is the integration of food recognition, visible freshness estimation, reference-based shelf-life guidance, persistent pantry history, deterministic urgency and conversational support in a single prototype. The work is an applied system and workflow contribution, not a newly invented CNN or a validated food-safety technology.

- Connects visual analysis to storage guidance and pantry records.
- Exposes an explainable Eat First tier system.
- Preserves saved context across sessions for follow-up assistance.
- Documents evaluation limits and refuses to promote a weaker/unreconciled model candidate.
- Defines recipes, PWA support, donation and measurable impact as future work.

26. If a judge remembers only eight things

1. Food waste is a household problem as well as a supply-chain issue; this project focuses on household decisions.
2. FoodFresh AI combines image analysis, storage references and saved pantry context in one workflow.
3. Recognition, freshness and shelf-life are distinct components with distinct limits.
4. Freshness output describes visible appearance, not certified safety.
5. Eat First is explainable deterministic prioritization, not a learned ranking model.
6. FreshoBuddy grounds “eat first” answers in pantry records instead of inventing pantry facts.
7. Pantry-aware recipes and NGO donation are future scope; no partnership or waste impact is claimed.
8. Model metrics are tied to named splits; real-world fixtures show errors and the V3 discrepancy remains disclosed.

27. Future vision



The sequence is a roadmap, not the current deployment state. Each step should follow data quality, usability and safety validation. A donation network in particular depends on real partner organizations and operational responsibility.

28. Main motto

Know Your Food. Understand what you have and what its visible condition suggests. **Save Your Food.** Use storage context and priority guidance to notice food before its estimated quality window closes. **Waste Less.** Consume or repurpose usable food-and, in a future verified network, redirect suitable surplus.

FoodFresh AI is not only about detecting food - it is about helping people decide what to do with it.

29. References

1. H. Mureşan and M. Oltean, "Fruit recognition from images using deep learning," *Acta Universitatis Sapientiae, Informatica*, 10(1), 26-42, 2018. DOI: [10.2478/ausi-2018-0002](https://doi.org/10.2478/ausi-2018-0002). Historical dataset snapshot; newer releases differ.
2. F. Risdin, H. Salauddin, T. A. Tufa and M. J. Hossain, "AgriFreshNET Freshness and Shelf-Life Image Dataset," Mendeley Data, V1, 30 Jul. 2026. DOI: [10.17632/42m5tb7yv9.1](https://doi.org/10.17632/42m5tb7yv9.1).
3. S. Liu et al., "Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection," arXiv:2303.05499, 2023 (revised 2024). arxiv.org/abs/2303.05499.
4. M. Tschannen et al., "SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features," arXiv:2502.14786, 2025. arxiv.org/abs/2502.14786.
5. J. Ma et al., "Applications of Computer Vision for Assessing Quality of Agri-food Products: A Review of Recent Research Advances," *Critical Reviews in Food Science and Nutrition*, 56(1), 113-127, 2016. DOI: [10.1080/10408398.2013.873885](https://doi.org/10.1080/10408398.2013.873885).
6. M. Palumbo et al., "Computer vision system based on conventional imaging for non-destructively evaluating quality attributes in fresh and packaged fruit and vegetables," *Postharvest Biology and Technology*, 200, 112332, 2023. DOI: [10.1016/j.postharvbio.2023.112332](https://doi.org/10.1016/j.postharvbio.2023.112332).
7. USDA Food Safety and Inspection Service, "FoodKeeper App," foodsafety.gov/keep-food-safe/foodkeeper-app; public FoodKeeper data catalog: catalog.data.gov/dataset/fsis-foodkeeper-data.
8. United Nations Environment Programme, *Food Waste Index Report 2024*, 2024. unep.org/resources/publication/food-waste-index-report-2024.
9. M. Oltean, "Fruits-360 dataset," official GitHub organization and versioned repository, checked 30 Sep. 2026: github.com/fruits-360/fruits-360-original-size. Dataset release counts change over time; license is CC BY-SA 4.0.
10. USDA FSIS, "FoodKeeper Data," Data.gov catalog entry, catalog.data.gov/dataset/fsis-foodkeeper-data.
11. I. Daud, "Raw Food Recognition Models: Single-Class and Multi-Label Classification," Hugging Face model card, 2024. External reported validation result only: huggingface.co/ibrahimdaud/raw-food-recognition-models.
12. Google AI for Developers, Gemini API documentation, ai.google.dev/gemini-api/docs.

External dataset/model claims above are attributed to their original sources. Project-specific scores and implementation status were checked against the local model configuration, evaluation reports, integration reports and API flow on 30 September 2026. Where the record was inconsistent, the limitation is disclosed rather than resolved by assumption.